

Sumukh Patel

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EDUCATION

- McGill University — M.Eng. Electrical Engineering, Applied Artificial Intelligence** Expected May 2027
Montreal, QC | GPA: 3.75/4.00
- Dhirubhai Ambani University (formerly DA-IICT) — B.Tech. (Honours), Information & Communication Technology** May 2025
Minor: Computational Science | CGPA: 8.7/10

RESEARCH & TECHNICAL EXPERIENCE

- Applied AI Research Project — LLM Trustworthiness & Hallucination** Jan 2026 – Present
McGill University, Department of Electrical & Computer Engineering | Supervisor: Prof. Di Wu
- Designed repeated experiments benchmarking multiple open- and closed-source LLMs on domain-specific question answering; measured category-level accuracy, invalid responses, consistency, and variance.
 - Built Python evaluation pipelines and performed focused error analysis on failure-prone categories; synthesized methodology, results, plots, limitations, and conclusions in an IEEE-format research report.
- Research Intern — Multimodal Hate Speech Detection** Jul 2024 – Aug 2024
Stiftung Universität Hildesheim, Germany | Information Science
- Fine-tuned and benchmarked LLM-based and multimodal models across four research datasets; evaluated performance using accuracy, F1, recall, and AUC.
 - Conducted literature review, data preprocessing, ablation/error analysis, and interpretation of experimental results to support ongoing research reporting.
- Research Project — Denoising Implicit Feedback for Recommender Systems** Dec 2023 – Jun 2025
Dhirubhai Ambani University | PyTorch / Recommender Systems
- Developed a novel label-flipping denoising method for implicit-feedback recommendation and evaluated it on multi-dataset experiments containing millions of interactions.
 - Built reproducible training/evaluation pipelines using Recall@N and NDCG@N; performed systematic hyperparameter, ablation, and datapoint-level analyses.
- Research Intern — Smart Energy Learning Centre** Dec 2024 – Jul 2025
Dhirubhai Ambani University, India
- Prepared and analyzed research datasets for renewable-energy applications; developed predictive/prescriptive ML analyses and investigated NLP/foundation-model approaches.

LEADERSHIP & ADDITIONAL EXPERIENCE

- Teaching Assistant — Software Design, Database Systems, Digital Logic** Aug 2025 – Present
McGill University
- Deliver tutorials, grade technical work, and communicate structured feedback; also serve as an Exam Coordinator supporting secure, time-sensitive examination operations.
- Google Live Projects — Open Source Project Contributor** Sep 2023 – Jan 2024
Google, Bengaluru, India | Embedded Systems
- Implemented Little Kernel thread-management features and analyzed ARM A-class processors; collaborated in a peer-review environment using testing and benchmarking.

TECHNICAL SKILLS

- Programming:** Python (advanced), Bash/Linux (advanced), C++ (intermediate), C (intermediate), SQL (intermediate), Java (working knowledge)
- ML / Data:** PyTorch, TensorFlow/Keras, scikit-learn, pandas, Transformers; LLMs, NLP, multimodal learning, recommender systems, computer vision, reinforcement learning
- Research:** Experimental design, statistical evaluation, ablation/error analysis, ranking metrics, classification metrics, technical writing, data visualization, Git/GitHub

RESEARCH CONTRIBUTION

- Co-author, shared-task overview:** "Overview of the HASOC Track at FIRE 2025: Abusive Meme Identification — Shadows Behind the Laughter," FIRE 2025 (ACM).